

PAC PERFORMANCE DATA

CONFIDENTIAL

Model Name: PLY-SM48EA4-TH,TH / PUY-SM48VKA4-TH,TH Indoor unit: PLY-SM48EA4-TH,TH
 Power supply: 1 Ø 50 Hz 230 V -Production date: 05-Mar-25 Line: P
 Comp.model: SVB220FBGM1T -Lot size: (53M00139 - 53M00194) Units 56
 Ref.weight: (R32): 1200 g. Outdoor unit: PUY-SM48VKA4-TH,TH
 C/T size: - mm. -Production date: 14-Mar-25 Line: W1
 LEV: UKV-25D251 -Lot size: (53P00116 - 53P00146) Units 30
 Reference design specification no.: MCP-23D011-PFD-B / MCP-23D009-PFD_G Period of test: 23 - 31 May '25

APPROVE (QA,Dept.Mgr.)	APPROVE (Dev. Test SM.)	CHECKE	RECORD
REPORT NO.	SAM-25-760-0006		

Reference Product standard: EGAT, TIS 2134-2553, TIS1529-2561, TIS812-2558

Reference Safety test standard TIS1529-2561, TIS812-2558

Reference Capacity and Performance test standard: ISO 5151: 2017, JIS-C9612, TIS 2134-2553

Reference EMC test standard: -

Reference Sound test standard: ISO 3744:2010, ISO 3745:2010

Reference Other test standard: -

1. Test result:

Classification of Evaluate Item		Specification and Tolerance		Qty	Judg	Data test result	
Test	Item	Condition					
Cooling • Heating Capacity and General Performance	Cooling capacity	Room temp.	Spec. 1 φ 50 Hz. 230 V.	Tolerance	1	OK	[PA-14B] S/N:(Indoor:53M00176 / Outdoor:53P00141)
		In DB27.0 °C, WB19.0 °C	Capacity= 14.100 kW/hr	Q ≥ 97.2 %			Capacity= 13.873 kW/hr (98.390%)
		Out DB35.0 °C, WB24.0 °C	Power input= 5400 W	P ≤ 110 %			Power input= 5563 W (103.019%)
			Current= 25.10 A	-			Current= 24.18 A (96.335%)
			EER= 2.61 W/W	EER ≥ 95.2 %			EER= 2.490 W/W (95.402%)
	Heating capacity	Room temp.	Spec. 1 φ 50 Hz. 230 V.	Tolerance	1	OK	[PA-14B] S/N:(Indoor:53M00176 / Outdoor:53P00141)
		In DB20.0 °C, WB - °C	Capacity (Spec) = 7.050 kW/hr	Q ≥ - %			Capacity (Spec) = 7.073 kW/hr (100.326%)
		Out DB7.0 °C, WB6.0 °C	Capacity (Rate / 2) = 6.937 kW/hr	Q ± 0.6937 kW			Capacity (Rate / 2) = 7.073 kW/hr (100.980%)
			Power input= 1370 W	Q ± 5% Actual rate			Power input= 1438 W (104.964%)
			EER = 5.15 W/W				EER = 4.920 W/W (95.534%)
	Cooling Overload	In 32.0 / 23.0, Out 43.0/26.0 °C	SEER (Label)= 19.06 Btu/Whr	SEER ≥ 95.2 %			SEER (Label)= 18.32 Btu/Whr (96.118%)
		In 32.0 / 23.0, Out 43.0/26.0 °C	EGAT Target random = 5	≥EGAT5 rank(SEER ≥ 14.00)			EGAT rating = 5★ (SEER = 18.32)
		In 21.0 / 15.0, Out 21.0/15 °C	Extension pipe length = 7.5 m.				Extension pipe length = 7.5 m.
		In 21.0 / 15.0, Out 21.0/15 °C					
		In 27.0 / - , Out 24.0 / 18.0 °C					
	Freezing disturbance	Low temp High Humidity ratio					
		In 20.0 / 12.0, Out 2.0 / 1.0 °C					
		In 27.0 / 24.3, Out 27.0/24.3 °C					
Noise • Vibration	Overall (SPL)	Cooling mode:	Indoor side:		1	OK	Indoor side: S/N 53M00176
		In DB27.0 °C, WB19.0 °C	Fan Rated Hi speed= 44 dBA (670 rpm)	not over + 4dBA			Fan Rated Hi speed= 48.7 dBA 664 rpm
		Out DB35.0 °C, WB24.0 °C	Fan Super Hi speed= dBA (rpm)	not over + 4dBA			
			Fan Hi speed= dBA (rpm)	not over + 4dBA			
			Fan Med speed= dBA (rpm)	not over + 4dBA			
	Abnormal noise	Heating mode:	Outdoor side:		1	OK	Outdoor side: S/N 53P00141
		In DB20.0 °C, WB - °C	Front side= 56 dBA (800 rpm)	not over + 4dBA			Front side= 56.5 dBA 796 rpm
		Out DB7.0 °C, WB6.0 °C	Back side= 56 dBA (800 rpm)	not over + 4dBA			Back side= 55.0 dBA 796 rpm
	Overall (SPL)	Cooling mode:	Feeling check no abnormality noise		1	OK	- Indoor start and stop have refrigerant noise and compressor noise level low.
		In DB27.0 °C, WB19.0 °C	- Pipe length 7.5 m.				- Outdoor start and stop have refrigerant noise and compressor noise level low.
		Out DB35.0 °C, WB24.0 °C					
EMC	Abnormal noise	Heating mode:	Feeling check no abnormality noise		1	OK	- Indoor start and stop have refrigerant noise and compressor noise level low.
		In DB20.0 °C, WB - °C	- Pipe length 5 m.				- Outdoor start and stop have refrigerant noise and compressor noise level low.
		Out DB7.0 °C, WB6.0 °C					
	Other test (Depend on change point)						
Safety	Impulse noise	Operating characteristic	Function operate are not abnormal When voltage supply £ 1.8 KV				
		Other test (Depend on change point)	Operate check confirm by IB				
	Insulation resistance	Dielectric Withstand Voltage	10 M Ω or more / 500 V	1	OK	100 M Ω	
			Cut off current 100 mV Basic Insulation 1000 V / 1 min	1	OK	PASS	
			Supplementary Insulation 1750 V / 1 min	1	OK	PASS	
			Reinforced Insulation 3000 V / 1 min	1	OK	PASS	
				1	OK	PASS	
	Earthing continuity	Leakage Current	25A between earthing and metal parts resistance not exceeding 0.1 W	1	OK	PASS	
			Not exceed 2 mA. per Kilowatt of Max power input. (12.08 mA)	1	OK	Max Peak : 1.21 mA. Cooler (N-E) Start	
			Power supply 402.8 V. 1.06 Times rated voltage				
Final Inspection	DT Packing Inspection	Accessories , IB , IM , Guarantee must be packing and check		1	OK	PASS	
		Function Test at Production Line	Can operate comply with model function	1	OK	PASS	
		PQI Packing Inspection	Accessories , IB , IM , Guarantee must be packing and check	1	OK	PASS	
	Other test (Depend on change point)						
	Other test (Depend on change point)						

2.Problem :

No.	Detail	Cause	Improvement	Judgement

COMMENT : Calculation half load heating capacity

Capacity of air conditioner		Investor SEER (Btu/h)					
$\left[\frac{(Q_{act} \text{ half} - Q_{half} \text{ rate}) \times 100}{Q_{act} \text{ rate}} \right] + 100$	13600 + 100	5	5★	5★	5★	5★	5★
	13873	17.06	18.56	20.06	21.56	23.06	24.56
		3000 W < Q ≤ 12000 W	16.04	17.16	18.28	19.39	20.51
$\left[\frac{(7073 - 6937) \times 100}{13873} \right] + 100$	0.980 + 100	12000 W < Q ≤ 16000 W	14.00	15.50	17.00	18.50	20.00
$\left[\frac{136 \times 100}{13873} \right] + 100$							